Solution File Description

Scope

This document describes the solution-result files in the solutions directory. Each file is tab-separated. The first column is the instance index, which follows the *m_n_k* convention used by the instance files.

- For exact-algorithm results, the instance index refers to files in instances/Instances-for-Exact-Algorithm.
- For approximation-algorithm results, the instance index refers to files in instances/Instances-for-Approximation-Algorithm.
- In Instance_*m_n_k*, *m* is the number of periods, *n* is the number of orders, and *k* is the replicate index for the same (*m*, *n*) setting.
- Running time is reported in seconds.

Result_E-ILP.txt

This file reports the results obtained by solving Model E-ILP via CPLEX with a time limit of 300 seconds.

Column	Meaning
Column 1: Instance	Instance index.
Column 2: upper_bound	The objective value of the solution obtained by solving Model E-ILP via CPLEX with a time limit of 300 seconds.
Column 3: Lower_bound	The lower bound generated by CPLEX with a time limit of 300 seconds.
Column 4: RunTime(s)	Running time.

Result-Approximation-Algorithm-Algorithm 4.txt

This file reports the results obtained by the approximation algorithm, Algorithm 4.

Column	Meaning
Column 1: Instance	Instance index.
Column 2: ObjValue	The objective value of the solution obtained by the approximation algorithm, Algorithm 4.
Column 3: RunTime(s)	Running time.

Result-Approximation-Scheme-Algorithm 6.txt

This file reports the results obtained by the approximation scheme, Algorithm 6. The file contains separate result blocks for different ε values. Lines such as $\varepsilon = 2$ and $\varepsilon = 1$ indicate the ε value used for the following block of instances.

Column	Meaning
Column 1: Instance	Instance index.
Column 2: ObjValue	The objective value of the solution obtained by the approximation scheme, Algorithm 6.
Column 3: RunTime(s)	Running time.

Result-Exact-DP-Pruned-Algorithm BDP.txt

This file reports the results obtained by the BDP algorithm with a time limit of 300 seconds. The columns also summarize state generation, pruning by lower bounds, and GUB updates by upper bounds.

Column	Meaning
Column 1: Instance	Instance index.
Column 2: OptimalValue	The objective value of the solution obtained by the BDP algorithm with a time limit of 300 seconds.
Column 3: RunTime(s)	Running time.
Column 4: TotalGenerateStates	The total number of states generated by the BDP algorithm.
Column 5: LB_RCprunningStates	The number of states pruned by the lower bound $LB^{RC}(w)$.
Column 6: LB_RCprunningRate(%)	The pruning rate of the lower bound $LB^{RC}(w)$.

Column 7: LB_PDprunningStates	The number of states pruned by the lower bound $LB^{PD}(w)$.
Column 8: LB_PDprunningRate(%)	The pruning rate of the lower bound $LB^{PD}(w)$.
Column 9: UB_BIupdateTimes	The GUB update times by the upper bound $UB^{BI}(w)$.
Column 10: UB_BIupdateRate(%)	The GUB update rate by the upper bound $UB^{BI}(w)$.
Column 11: UB_BSupdateTimes	The GUB update times by the upper bound $UB^{BS}(w)$.
Column 12: UB_BSupdateRate(%)	The GUB update rate by the upper bound $UB^{BS}(w)$.

Result-Exact-DP-Unpruned-Algorithm 3.txt

This file reports the results obtained by the exact algorithm, Algorithm 3, with a time limit of 300 seconds.

Column	Meaning
Column 1: Instance	Instance index.
Column 2: OptimalValue	The objective value of the solution obtained by the exact algorithm, Algorithm 3, with a time limit of 300 seconds.
Column 3: RunTime(s)	Running time.
Column 4: TotalGenerateStates	The total number of states generated by Algorithm 3.